Quantifying Classifier Utility under Local Differential Privacy

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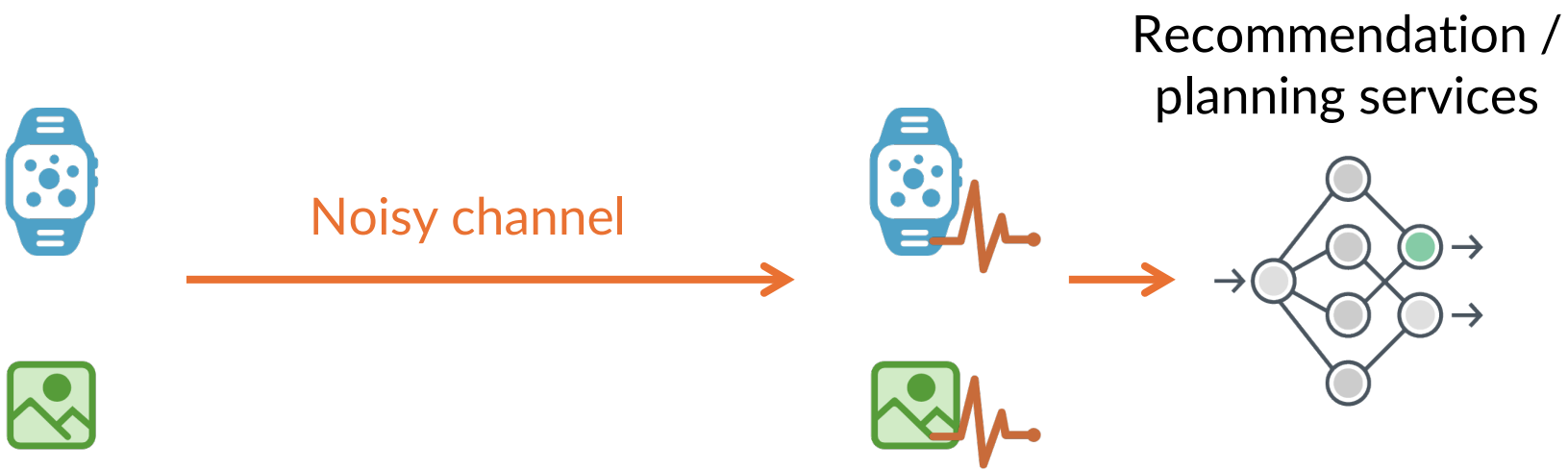
Classifier Utility under Noisy Inputs

- Input data for classifiers may be noisy



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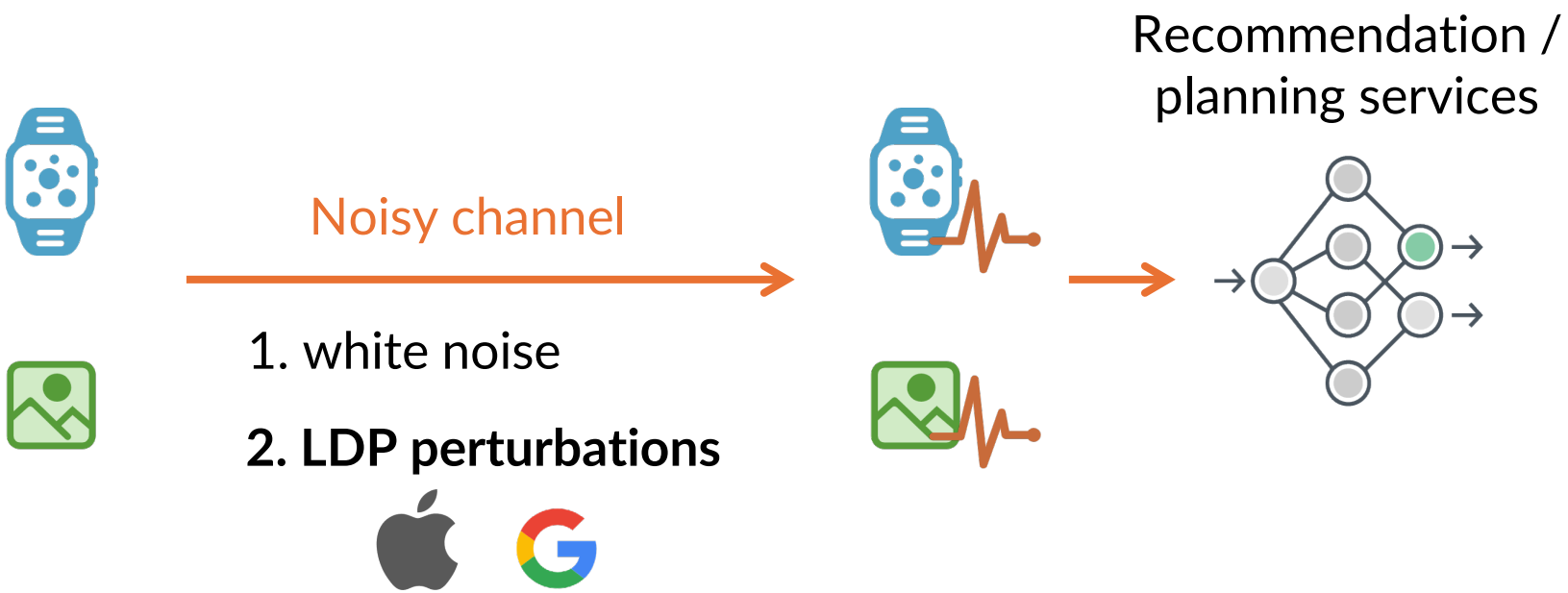

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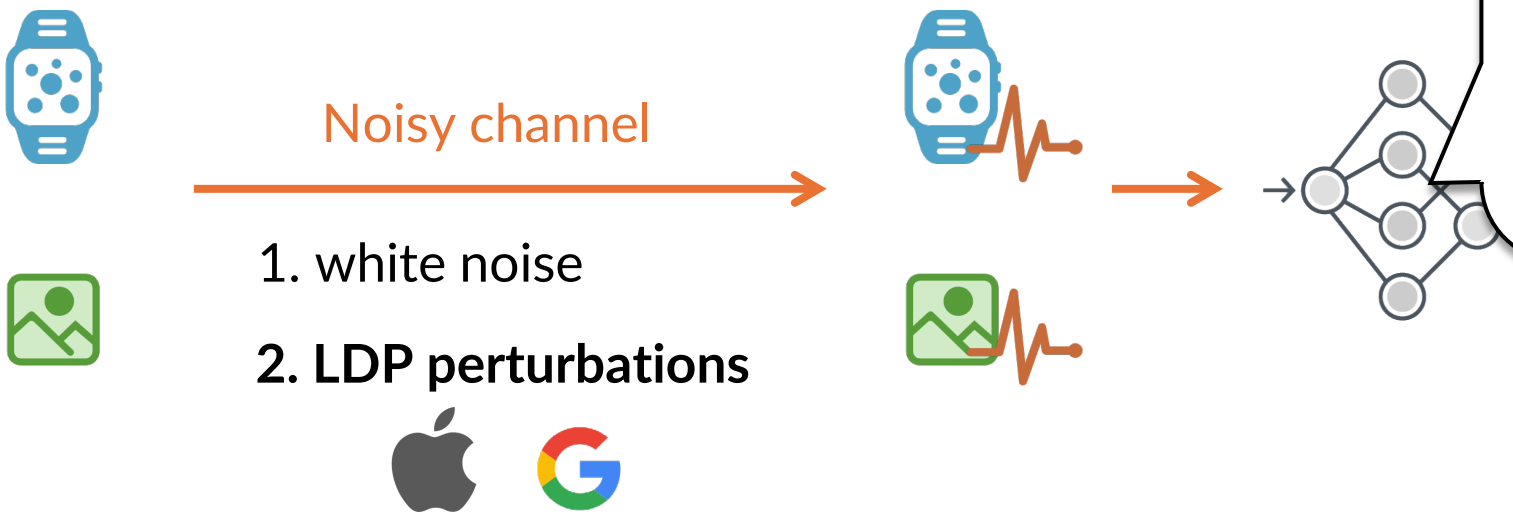


What's the utility of the trained classifier under a noisy channel, e.g. LDP perturbation?

Classifier Utility under Noisy Inputs

- Input data for classifiers may be noisy

What's the utility of the trained classifier under a noisy channel, e.g. LDP perturbation?

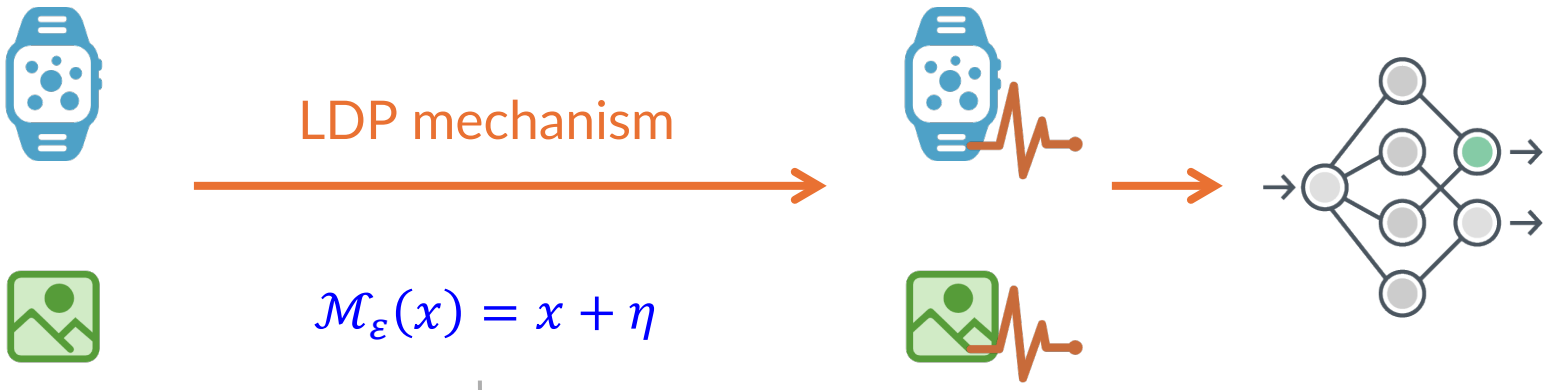
- Q: How can classifier designers/users know the classifier's accuracy under LDP-perturbed data?

For an LDP-friendly classifier

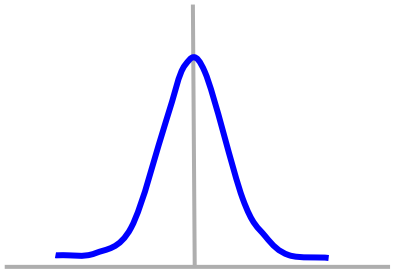
For a better privacy-utility balance

Empirical Classifier Utility under LDP-Data

- Empirical approach: Sample and then test



$$\mathcal{M}_\varepsilon(x) = x + \eta$$



$$\eta \sim \mathcal{N}(0, \sigma^2)$$

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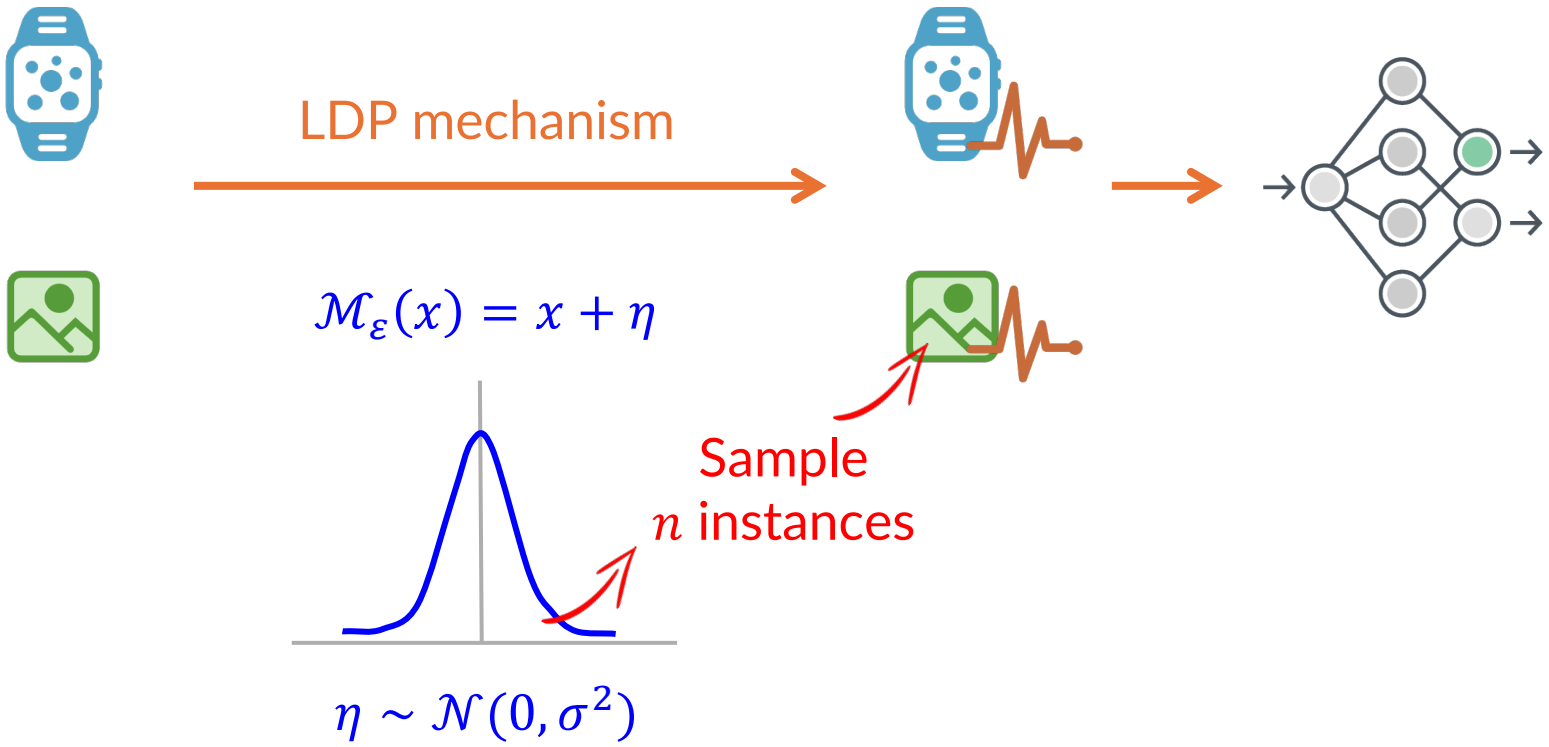

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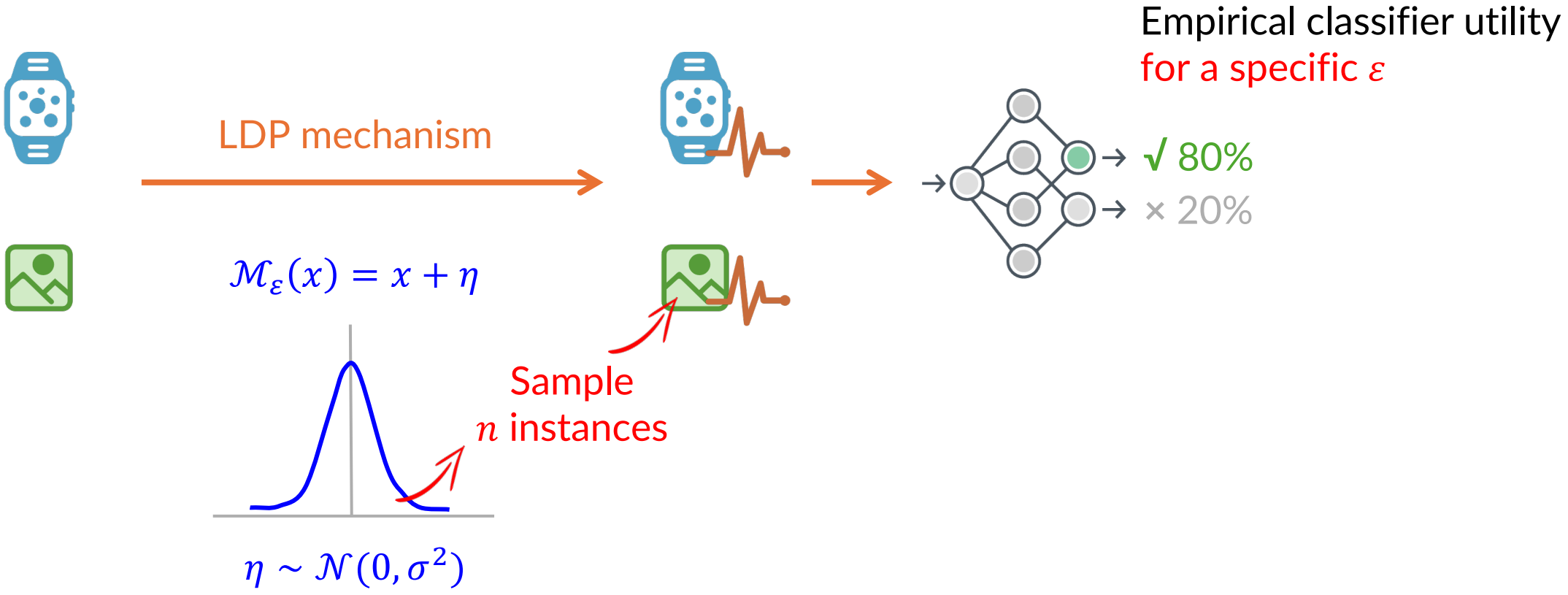

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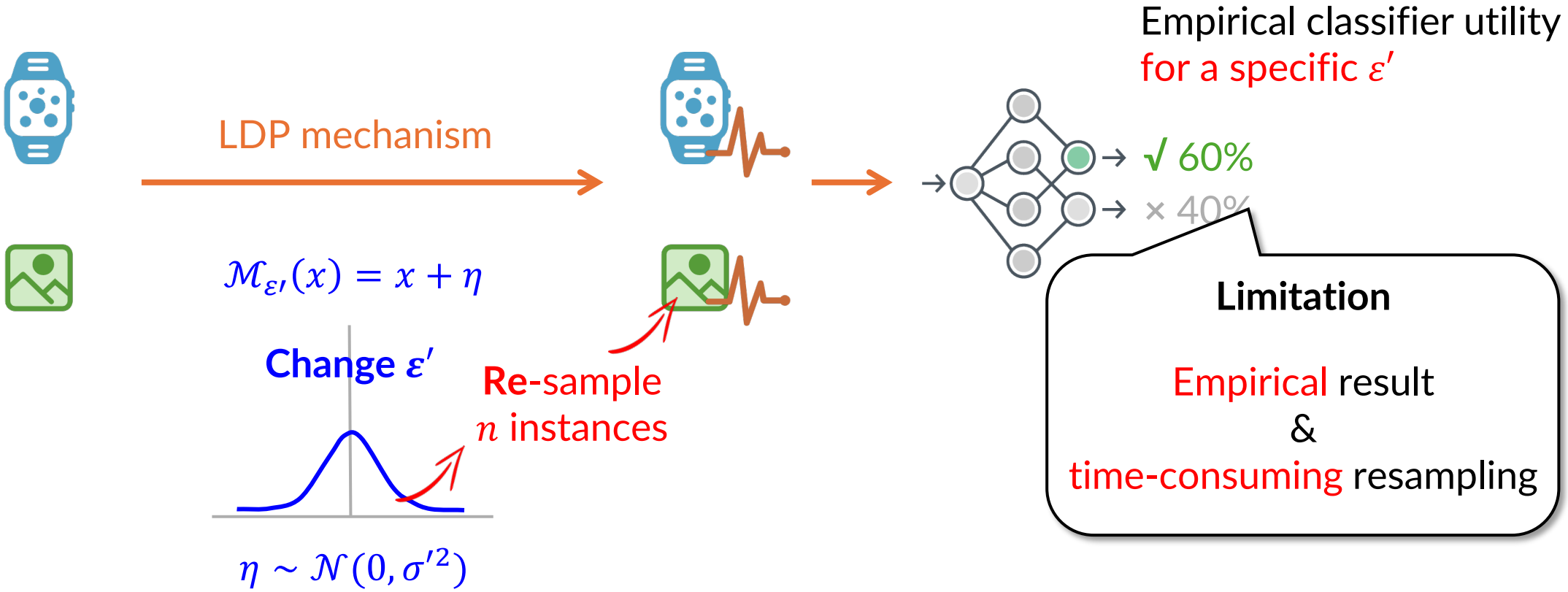

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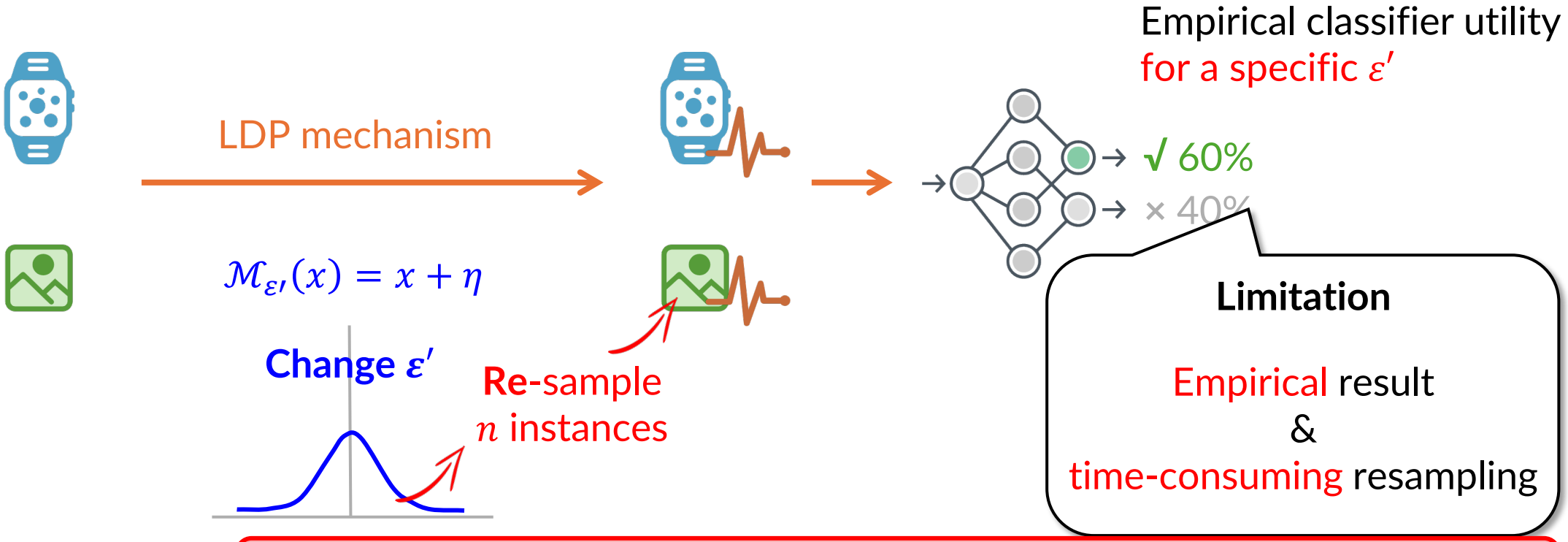
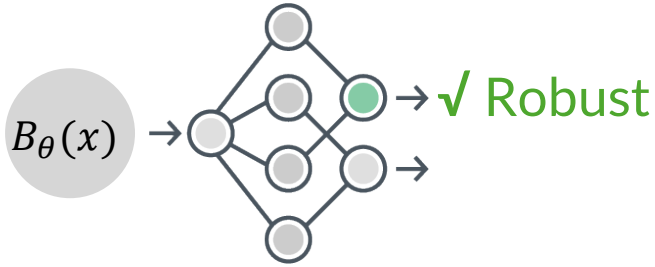


Lacks of an **analytical view** of classifier utility under LDP data

Empirical Utility → Analytical Utility

- Analytical approach: connecting LDP with robustness

$B_\theta(x)$ is robust



Robustness:
A θ-ball around x that doesn't change classification

Empirical Utility → Analytical Utility

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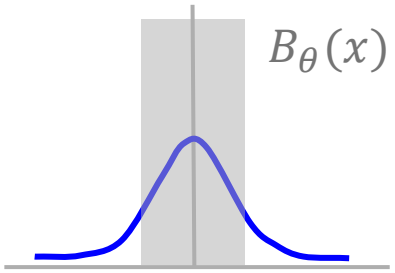
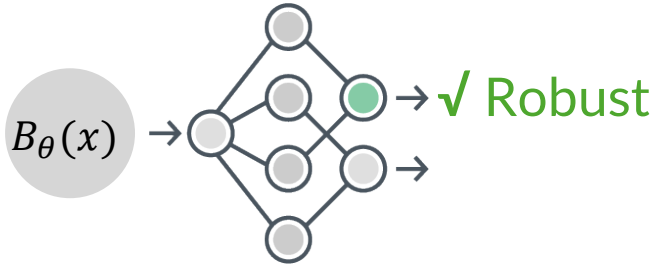
$\mathcal{M}_\varepsilon(x)$ concentrates in $B_\theta(x)$



Concentration:
Probability of falling into a region
$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)] \coloneqq p(\varepsilon, \theta)$

$B_\theta(x)$ is robust



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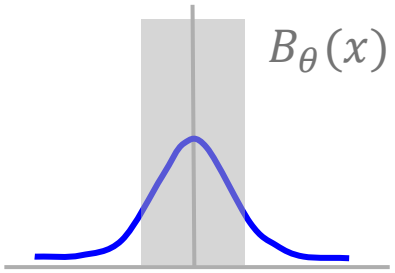

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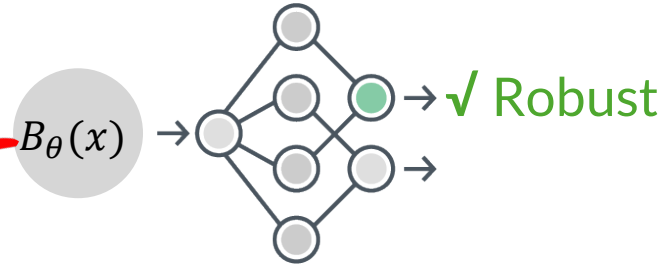

Concentration:
Probability of falling into a region
$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)] := p(\varepsilon, \theta)$

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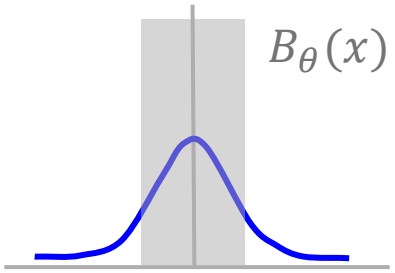

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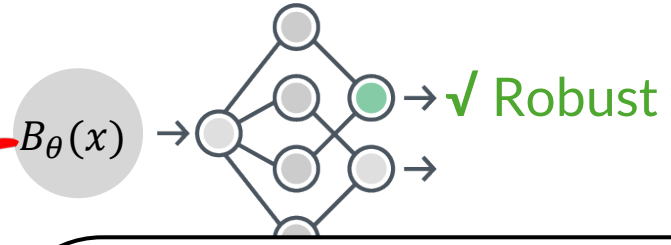


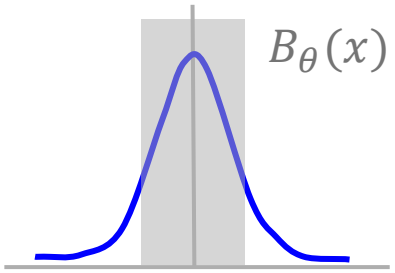
Analytical classifier utility

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.

Empirical Utility → Analytical Utility

- Analytical approach: connecting LDP with robustness

$\mathcal{M}_{\varepsilon'}(x)$ concentrates in $B_\theta(x)$



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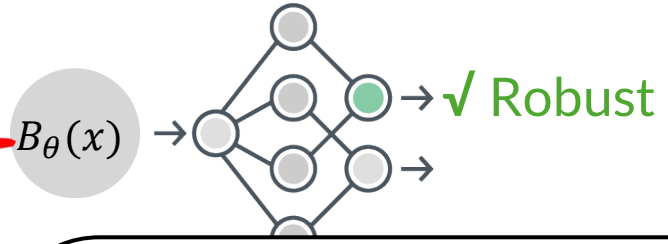


Analytical classifier utility

With probability at least $p(\varepsilon', \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_{\varepsilon'}(x)$.

Analytical & systematic view for any ε without resampling

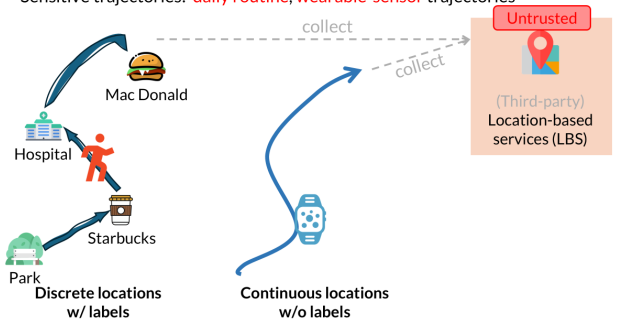
One-Dimension Example

- Input data for classifiers may be noisy

TraCS: Trajectory Collection in Continuous Space under LDP

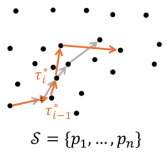
Trajectory Collection

- Sensitive trajectories: daily routine, wearable-sensor trajectories



LDP-fy a Trajectory in A **Discrete Space**

- Discrete location spaces: Point of interests or discretized ce



Limitations

1. **Efficiency**
   - each sample costs $\mathcal{O}(n)$
2. **Trajectory Utility**
   - decreases as n increases
3. **Applicability**
   - cannot apply to continuous $\mathcal{S}$

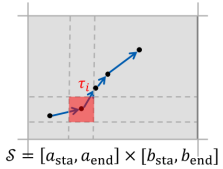
- Exponential mechanism:

$$\Pr[\mathcal{M}_{\exp}(\tau) = \tau^*] = \frac{\exp(\varepsilon d(\tau, \tau^*))}{\sum_{\tau' \in \mathcal{S}} \exp(\varepsilon d(\tau, \tau'))}$$

← Pairwise distance
← Sum of distance

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



Simple sampling: (satisfying LDP for $\mathcal{S}$)

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_\varepsilon$

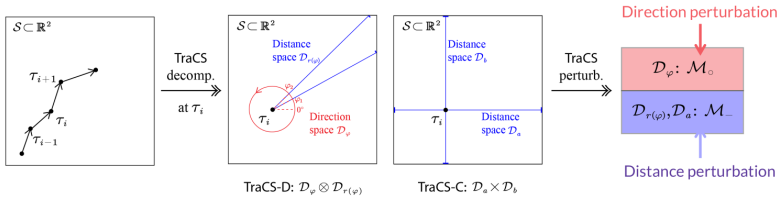
$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_\varepsilon / e^\varepsilon$

- Benefits:
  1. **Efficiency:** $\mathcal{O}(1)$ sampling complexity
  2. **Trajectory utility:** "n-independent"
  3. **Applicability:** Both continuous spaces & discrete spaces

TraCS: Trajectory Collection in Continuous Spaces

- TraCS-D: direction-distance perturbation
- TraCS-C: coordinates perturbation

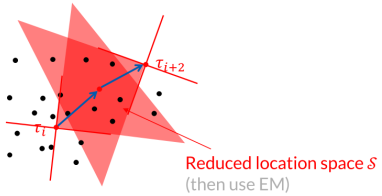
- **Key idea:** decomposes $\mathcal{S}$ into two subspaces → design $\mathcal{M}$ for each subspace



Evaluations

- Claims
  - continuous spaces: better utility than naïve baselines, e.g. planar Laplace + truncation
  - discrete spaces: better efficiency and utility than existing methods, ATP*, NGram**, L-SRR***

- **ATP:** direction perturbation
  1. divide direction sectors, e.g. $k = 4$
  2. perturb sector
  3. $\mathcal{M}_{\exp}(\tau)$ on reduced $\mathcal{S}$



* Trajectory Data Collection with Local Differential Privacy, VLDB'23
** Real-World Trajectory Sharing with Local Differential Privacy, VLDB'21
*** L-SRR: Local Differential Privacy for Location-Based Services with Staircase Randomized Response, CCS'22

Evaluations – Discrete Spaces

- TKY and CHI datasets
  - TraCS + rounding-to-the-nearest

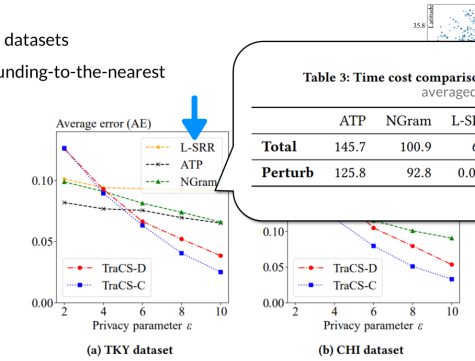


Table 3: Time cost comparison (in milliseconds). averaged

	ATP	NGram	L-SRR	TraCS-D	TraCS-C
Total	145.7	100.9	6.2	0.06	0.05
Perturb	125.8	92.8	0.003	0.018	0.003

Thank you!